EFFECTS OF SIGNAGE AND FLOOR PLAN CONFIGURATION ON WAYFINDING ACCURACY

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ABSTRACT: Signage is commonly employed to enhance wayfinding efficiency, especially in buildings with complex floor plan configurations. This study examines the influence of floor plan complexity and several types of signage on wayfinding within a series of buildings on a university campus. The study used a 5 x 3 factorial experimental design. The first factor, complexity of floor plan configuration, is defined through five alternatives. The second factor, signage, has three conditions: no signage, textual signage, or graphic signage. The results show that as floor plan complexity increases, wayfinding performance decreases. Graphic signage produced the greatest rate of travel in all settings, but textual signage was the most effective in reducing wayfinding errors, such as wrong turns and backtracking. Overall, the addition of signage resulted in a 13% increase in rate of travel, a 50% decrease in wrong turns, and a 62% decrease in backtracking across the five settings. However, plan configuration was found to exert a significant influence regardless of signage, because the wayfinding performance of participants with access to signage in the most complex settings remained equivalent to, or significantly poorer than, those in the simplest settings with no signage.

People rely on numerous types of environmental information to find their way within buildings. Weisman (1981) developed four classes of environmental variables thought to influence wayfinding: (a) visual access to familiar cues or landmarks within or exterior to a building; (b) the degree of architectural differentiation between different areas of a building that can aid recall and orientation; (c) the use of signs and room numbers to provide identification or directional information; and (d) plan configuration, which can influence the ease with which one can comprehend the overall layout of the building. Related to floor

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plan configuration, research suggests that the quantity of (Best, 1970) and complexity of relations between choice points (Hillier, Hanson, & Peponis, 1984; O'Neill, 1991; Peponis, Zimring, & Choi, 1990) such as hallway intersections in buildings, may influence wayfinding.

Of these variables, a number of studies suggest that the complexity of floor plan configuration is a primary influence on wayfinding performance. Best (1970) found a positive relationship between the number of choice points within a floor plan and wayfinding difficulty. Weisman (1981) found that students reported being lost less frequently in university buildings whose floor plans were judged "simpler" and more "legible." This effect remained even for people who were very familiar with the buildings. Weisman found that ratings of floor plan complexity predicted 56% of the variance in reported disorientation in the actual buildings. Bronzaft and Dobrow (1984) suggest that simplicity and regularity of floor plans aid people in learning about the layout of a space. Nichols, Canete, and Tuladhar (in press) suggest that the primary cause of wayfinding difficulty in transportation centers is the complexity of configuration of hallways and number of choice points within them. O'Neill (1991) found that with even incremental increases in floor plan complexity, people have significantly greater problems understanding spatial layout, and reduced wayfinding performance. Beaumont, Gray, Moore, and Robinson (1984) interviewed building users and found that floor plan layout was equal in importance to other architectural features, such as signage.

Signage is commonly employed to compensate for the complex floor plan layouts of settings such as subways, hospitals, and large governmental buildings. These are environments in which wayfinding is a chronic problem. There are several different types of signage, including identification (room numbers or labels) and directional (such as "you-are-here" maps, arrows, or text directions; Carpman, Grant, & Simmons, 1984; Passini, 1980).

Best (1970) found that signage placed at decision points in buildings improved wayfinding performance. Corlett, Manenica, and Bishop (1972) applied Best's (1970) principles for signage placement to the renovation of a signage system in a university building, and found that people took significantly less time to find their destinations after the signs were simplified and moved to decision points. Levine (1982) and Levine, Marchon, and Hanley (1984) found that the orientation of you-are-here maps significantly influenced the ability of people to successfully complete wayfinding tasks. Wener and Kaminoff (1983) found that the addition of signage to a visitors' reception area in a correctional facility reduced visitor stress and significantly improved visitor wayfinding performance. Carpman et al. (1984) found that with an increase in the number of signs in a hospital hallway, wayfinding performance decreased. They also found that adequate signage reduced perceived stress among hospital visitors.

Other research suggests that signage is less than effective in supporting the wayfinding task. Carpman, Grant, and Simmons (1985) found that the wayfinding behavior of people initially entering a hospital was influenced more by visual access to the destination than by the available signage. Weisman (1987) found that only 18% of nursing home residents mentioned the use of signage as a strategy for wayfinding. The remaining 82% mentioned using other architectural features as cues for orientation. Seidel (1983) found that 76% of people who had difficulty wayfinding in a large metropolitan airport had trouble understanding the signage, and 30% of the sample felt there were too many signs. In a review of almost 400 fires, Bryan (1982) found that in buildings with signage, less than 8% of people fleeing fires reported relying on signage to find their way to exits. These studies suggest that in some settings signage enhances wayfinding and reduces confusion and stress, but in other cases it may be ignored completely. Therefore, although signage is commonly employed as a corrective measure in complex buildings, the research on signage is equivocal as to its effectiveness.

Most wayfinding research has focused on the influence of individual architectural variables on wayfinding (see Weisman, 1985). However, a few studies have recognized the need for a multivariate research approach. Using a computer graphic simulation of movement through an urban street setting, Evans, Skorpanich, Garling, Bryant, and Bresolin (1984) manipulated path configuration and different types of visual landmarks to determine the impact of these variables on the accuracy of the cognitive map. They found that landmarks aided place recognition but not route memory. The costs associated with the simulation reduced the number of experimental conditions, so there was no way to determine interactions between the landmark and street layout conditions. Because the researchers used a noninteractive computer simulation of an imaginary setting, there was no way to assess actual wayfinding performance.

Garling, Lindberg, and Mantyla (1983) examined the influence of visual access, familiarity, and availability of a floor plan map, on measures of orientation within a university building. The setting afforded a high level of visual access. A low visual access condition was created by artificially restricting participants' vision. All participants were given four tours of the building and asked to make distance and direction estimates to various locations within the setting after each trial. The low visual access group learned significantly more slowly than people with visual access. When given access to a floor plan map, the low visual access group learned as quickly as the visual access group.

Some parallels may be drawn between the variables used in these two studies and the architectural variables of visual access, floor plan configuration, and signage described by Weisman (1981). Evans et al.'s (1984) study used a computer simulation to model grid and nongrid street layouts. The effects of manipulating street layout may be analogous to those observed with different floor plan configurations. Their findings regarding landmarks could be generalized to landmarks found within and exterior to buildings. Garling et al.'s (1983) study simulated the effects of buildings designed with high and low visual access. Their use of floor plan maps parallels the application of schematized floor plans, such as you-are-here maps. Their findings suggest that you-are-here maps be able to overcome the negative effects of buildings designed with low visual access.

Unfortunately, neither of these studies incorporated observations of actual wayfinding behavior as an outcome measure.

This review suggests that complexity of floor plan form negatively influences wayfinding performance. Although the literature is equivocal on the usefulness of signage, signs are frequently used in an attempt to mitigate wayfinding problems. An interaction between signage and plan configuration may be partly responsible for the lack of consistent findings related to the influence of signage on wayfinding. However, because of the difficulty in assessing the effects of multiple variables in field studies, little research has investigated the influence of more than one architectural variable on wayfinding.

Therefore, the present study has several objectives: (a) to develop a group of settings that vary in floor plan complexity in order to form an ecologically valid sample of environments; (b) to examine the influence of floor plan complexity on observed wayfinding performance; (c) to investigate the influence of different types of signage on wayfinding performance; and (d) to determine whether there is an interaction between floor plan complexity and the presence and type of signage on wayfinding. It is hypothesized that wayfinding performance will decrease as floor plan complexity increases, and that wayfinding performance will be enhanced in settings with signage. Finally, it is hypothesized that there will be an interaction between the effects of floor plan complexity and signage on wayfinding. The literature suggests that signage may not be able to compensate for the influence of plan configuration in extremely complex settings.

EMPIRICAL STUDY

SELECTION OF ENVIRONMENTAL SETTINGS

Weisman (1981) used a card-sorting technique to assess the "goodness" of form of floor plan configurations. Following on Weisman, two card-sorting tasks were used to select the five settings to be used in the present study. These settings are thought to represent a range of simple to complex floor plans. Portions of the floor plans of a number of buildings on the University of Wisconsin, Milwaukee campus were schematized, resulting in 17 distinct shapes. An additional 8 shapes were derived by manipulating several of the legitimate floor plan forms. This was done to produce wider variability in form shape within a set of 25 cards containing these drawings that were presented to participants for evaluation (see Figure 1).

Two independent groups of judges (15 per group) were asked to rate the shapes along a 5-point Likert-type scale on one of two criteria: (a) simplicity of each diagram; or (b) legibility (ease of understanding of the floor plan) of a building with such a floor plan. The raters responding to the first set of criteria were not told the drawings represented floor plans, rather, they were told the drawings were nothing more than abstract two dimensional shapes. The raters in the second group were not told of the origin of the buildings from which the shapes were derived.

The weighted mean of the rating for each form was then plotted for each of the 25 cards in both the simplicity and legibility conditions. A correlation analysis using Spearman's Rho (r) found a .91 ($p < .001$) correlation between simplicity and legibility ratings. Because of this strong correlation, the weighted raw scores for both evaluations were combined and divided by 5 to produce an interval from which forms could be selected. One shape that was a "best fit" was rejected because it was a manipulation, not an existing floor plan. In this case, the closest shape representing an existing floor plan was selected. Figure 2 illustrates the schematized floor plans of the settings used in this study. The numbers above each shape are used to refer to individual settings throughout this article.

To further evaluate the range of complexity of the forms, they were analyzed using a measure of topological floor plan complexity developed by O'Neill (1991). This measure is called Inter-Connection Density (ICD), and is an overall measure of the average number of possible paths one can take from any choice point (such a hallway intersection or corridor turn) within a floor

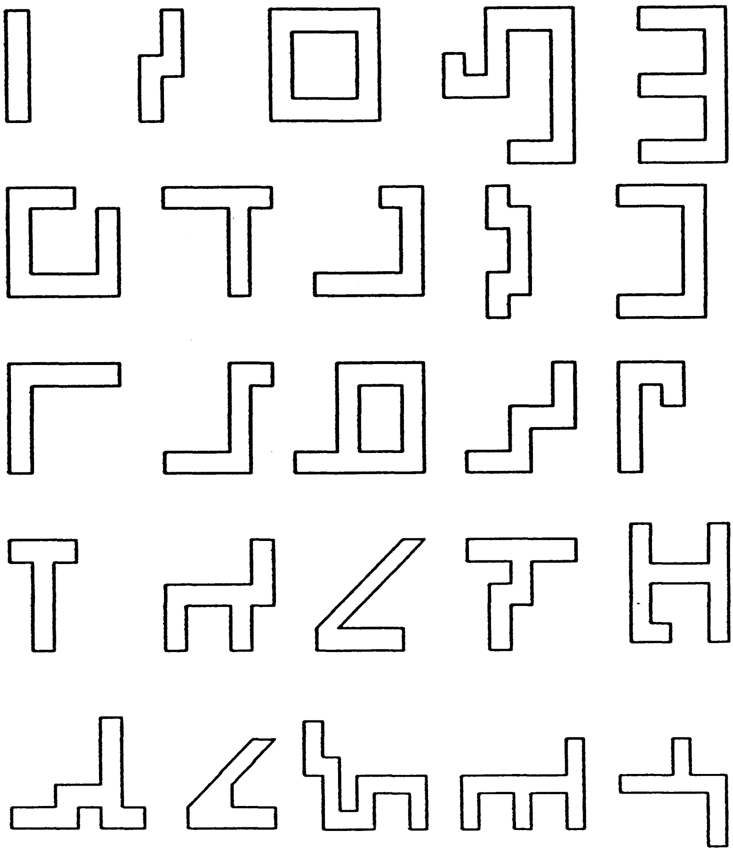

Figure 1: 25 Diagrams Used in Card-Sorting Tasks

plan. O'Neill found that as average ICD increases, cognitive mapping accuracy and wayfinding performance decrease.

In the present study, the average ICD for each of the 25 forms was calculated. The ICD values for the five settings selected are shown under each setting number (see Figure 2). These range from an average ICD of 1.00 for Setting 1, to 1.75 for Setting 5. This objective analysis of topological complexity supports the raters' evaluations on the two dimensions. With the exception of Setting 3, the forms show a gradual increase in complexity

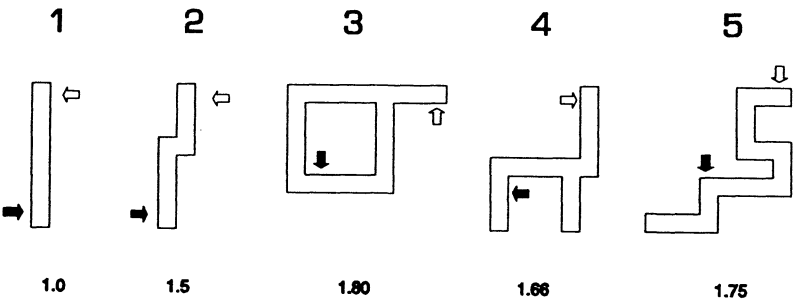

Figure 2: Schematic Floor Plans of Settings Selected by Raters

(as measured by ICD) from Settings 1 to 5. Setting 3 has an average ICD of 1.80, suggesting that it should have been selected as the most complex shape of this group. However, Setting 3 is also the only symmetrical form of the group. Across all 25 forms, a significant correlation ($r = .78, p < .01$) was found between rated complexity and average ICD value, and between rated legibility and average ICD value ($r = .73, p < .01$). Thus ICD is a strong predictor of raters' evaluations of form complexity and floor plan legibility, accounting for 61% of the variance in complexity ratings and 53% of the variance in legibility ratings.

CONTENT ANALYSIS OF CARD SORTS

After completing the card-sorting task, participants were asked to describe any additional criteria they employed in sorting the cards. A content analysis was performed to organize their responses into discrete categories. The responses to this query are summarized in Table 1.

This table lists the categories in order of frequency of mention. "Symmetry" was the most often named construct in the simplicity rating task (32%), and "ease of description" was mentioned by 34% of respondents in the legibility rating task. Raters indicated that symmetry reduces the complexity of a form. A symmetrical form contains redundant information, making it easier to describe (Levi, 1974). This may explain why Set-

TABLE 1
Content Analysis of Raters' Subjective Criteria for Card-Sorting Task (in percentages)

Simplicity Task	*Frequency Mentioned*	*Legibility Task*	*Frequency Mentioned*
Named construct		Named construct	
Symmetry	32	Ease of description	34
Intersections	25	Symmetry	20
Familiarity	20	Number of turns	20
Number of segments	12	Enclosure of space	17
Miscellaneous	11	Miscellaneous	9
	100		100

ting 3 was rated at the midpoint of the five settings, despite its relatively higher ICD value.

SIGNAGE

Both textual and graphic signage were used in this study. Signs were created by drawing symbols or text on 8 in. × 16 in. white foam core board and applying a plastic laminate over the surface to give each sign a finished look. The textual signage bore one of three phrases: "Turn Left Here," "Turn Right Here," or "Destination ahead." Each graphic sign contained an arrow indicating direction. The signage was visually distinct from the legitimate signage installed in each building so there was no confusion over which signage related to the experimental task. This signage was mounted near each decision point at a height of 5 ft 6 in. from the floor.

METHOD

RESEARCH DESIGN

A 5 × 3 factorial between-subjects experimental design was used. The first factor is configuration of floor plan form as defined by one of five alternatives. The second factor is signage type: no signage, textual signage, or graphic signage. The dependent variable, accuracy of wayfinding, is defined both as a function of rate of travel in reaching the predetermined goal point and the number of wayfinding errors (backtracking, wrong turns, stopping and looking) that occur while completing the task.

PARTICIPANTS

The 55 volunteers (21 females and 34 males) who participated in this study were primarily undergraduate students. They were randomly assigned to one of the five settings. To model the typical "occasional" building user, some familiarity with the building was desired. Each participant had been in the building an average of 3.6 times (standard deviation = 1.3). There was no significant difference in average amount of experience within the experimental settings between groups.

PROCEDURE

Each participant was brought to the start position (indicated by the black arrow on each floor plan in Figure 2) and told that the destination was a door with a red "X" taped on it (the goal location is shown by the unshaded arrow on each floor plan in Figure 2). The distance between start and destination in each setting was equivalent. This artificial destination was created for two reasons. Because every participant had some knowledge of the setting, it prevented prior knowledge of specific locations from confounding the task. Further, it forced participants to rely on knowledge of floor plan layout during the search, because they could not rely on existing signage or architectural cues to guide them to the artificial destination.

Thus each participant had some knowledge of floor plan layout, but did not know the specific location of the destination. This replicates a situation where a casual user has some knowledge of building layout but must locate a novel destination.

The efficiency of such a search is influenced by the user's knowledge, or cognitive map, of the form of the floor plan layout (Kaplan & Kaplan, 1982). The accuracy of the cognitive map is influenced by the complexity of the configuration of the floor plan (Weisman, 1981). The experimenter followed and marked each participant's wayfinding behavior on a coding sheet. To minimize extraneous influences such as the presence of other people in the hallways, the wayfinding tasks were conducted during times when classes were not in session.

MEASURES

Wayfinding performance was measured through four dependent variables: rate of travel (measured in feet per second), backtracking, stopping and looking, and wrong turns. Similar measures of wayfinding behavior have been used in other studies (Carpman et al., 1984; Weisman, O'Neill, & Doll, 1987; O'Neill, 1986; O'Neill, 1991, O'Neill, in press). Backtracking is defined as a retracing of a path in the opposite direction of how it was first traveled. Stopping and looking occurs when the participant hesitates or stops to look at signage or for orientation. A wrong turn is scored when the participant makes a turn in a direction that is incongruent with the most efficient completion of the wayfinding task. These measures were scored on an ordinal Likert-type scale to assess the magnitude of the behavior. A wrong turn, for example, could be measured from a 1 (participant turns and takes one step in wrong direction) to 5 (participant continues to end of corridor segment of wrong direction). These fine-grained measures were developed because people often begin to commit a wayfinding error or behavior and then quickly correct themselves. Such nuances can be missed during observations that are based on a dichotomous evaluation of the occurrence of a given behavior. Participants were not told that they were being evaluated on these measures, and there were no time or other constraints on the task.

TABLE 2
Summary of Significance and Direction of Means for Wayfinding Measures

	Setting
Rate of travel	5 < 2 < 4 < 1 < 3*
Backtracking	1 < 2 < 3 < 4 < 5***
Stopping and looking	12 < 345*
Wrong turns	123 < 45**

*$p < .0001$; **$p < .01$; ***$p < .05$.

RESULTS AND DISCUSSION

FLOOR PLAN COMPLEXITY

To determine the effects of floor plan configuration on wayfinding performance, the main effects for this variable were computed from a 2-way ANOVA (Setting × Type of Signage) on each of the dependent variables. Duncan's test was calculated to determine significant differences for each dependent measure between the individual settings. The findings of these analyses relating to the effects of floor plan configuration are summarized in Table 2. In this table, the "<" (less than sign) indicates the direction of the difference between significantly different means. There is a significant difference in rate of travel between each of the settings ($F[4, 54] = 12.14, p < .0001$) although not entirely in the expected ordering. As anticipated, rate of travel was lowest in Setting 5, but was highest in Setting 3 (see Table 2). There is a significant difference in backtracking behavior across the five settings ($F[4, 54] = 3.60, p < .05$). Duncan's test showed a significant difference in backtracking between each of the settings, in the anticipated ordering (see Table 2). Setting 1 elicited the lowest amount of backtracking, and Setting 5 the highest. There is a significant difference in stopping-and-looking behavior across the five settings ($F[4, 54] = 49.34, p < .01$) Duncan's test indicates that there is a significant difference in stopping and looking between two groups of settings, Settings 1, 2 and 3, 4, 5 (see Table 2). There is less stopping-and-looking behavior in the two simplest settings than in the three most complex settings. There is a significant difference in the amount of wrong turns across the five environments ($F[4, 54] = 14.72$, $p < .01$). Duncan's test detected significant differences in wrong turn behavior between two groups of environments, Settings 1, 2, 3 and Settings 4, 5.

The results indicate that there is not always an increase in wayfinding error for every incremental increase in plan complexity, but there is overall an increase in error as plan complexity increases. Plan complexity did not exert the same influence on rate of travel as for the other dependent measures. It was predicted that Setting 5 would have the lowest rate of travel, and Setting 1 the highest. Instead, Setting 3 had the highest rate of travel (see Table 2). One explanation for the high rate of travel in Setting 3 is that this environment has the only symmetrical plan layout in the group. This symmetrical form may be easier for people to understand and use despite its relatively higher topological complexity (for average ICD values see Figure 2), thus resulting in more confidence in travel and greater walking speed. Further, Levi (1974) suggests that symmetrical forms are less complex because they contain redundant information.

SIGNAGE

To determine the effects of signage on wayfinding performance, the main effects for this variable were computed from a 2-way ANOVA (Setting × Type of Signage) for each of the dependent measures. Duncan's test was calculated to determine significant differences for each dependent measure between the individual settings. The findings of these analyses relating to the effects of signage are summarized in Table 3. In this table, the "<" (less than sign) indicates the direction of the difference between significantly different means.

The dependent measures were evaluated for three signage conditions: textual signage, graphic signage, and no signage (see Table 3). The signage conditions reflect average performance across all settings. There is a significant difference in

TABLE 3
Summary of Effects of Signage on Dependent Measures

	Signage Condition
Rate of travel	3 < 1 < 2**
Backtracking	1 < 2 < 3*
Stopping and looking	n.s.
Wrong turns	1 < 2 < 3**

NOTE: 1 = textual signage, 2 = graphic signage, 3 = no signage; n.s. = not significant.
*$p < .0001$; **$p < .01$.

rate of travel between the three signage conditions ($F[4, 54] = 9.94$, $p < .01$). Duncan's test indicates that across settings, rate of travel is lowest when there is no signage, significantly higher when textual signage is used, and highest when graphic signage is used (see Table 3).

There is a significant difference between the signage conditions on backtracking behavior ($F[4, 54] = 13.90$, $p < .0001$). Duncan's test indicates significant differences between all three conditions (see Table 3). Textual signage produced the lowest amount of backtracking, graphic signage a greater amount of backtracking, and the most backtracking occurred when there was no signage present.

There was no significant difference in stopping-and-looking behavior between the signage conditions (see Table 3). It was thought that people would tend to hesitate more in settings with signage than in the nonsignage conditions. In fact, the opposite was true. The signage may have been simple enough for people to read as they walked, and hesitations may have been due to disorientation and lack of information in the nonsignage condition.

There is a significant difference between the signage conditions for number of wrong turns made ($F[4, 54] = 21.61$, $p < .01$). Duncan's test indicates significant differences between all three conditions (see Table 3). Participants using textual signage made the fewest wrong turns. People who had access to graphic signage made significantly more wrong turns, and people with no access to signage made the greatest number of wrong turns.

These results suggest that when time is not a constraint, textual signage is generally the most effective means of reducing major wayfinding errors, such as backtracking and wrong turns. Perhaps textual signage produces the greatest accuracy because it provides the most explicit information to users. It probably takes longer to process textual information, hence the sacrifice in rate of travel with this type of signage. Graphic symbols are less effective, but still better than no signage at all. Graphic signage may be most appropriate for conditions in which rate of travel is the top priority and accuracy is less important. To enhance rate of travel, graphic symbols might be used as "reassurance" signage between major signage locations.

FLOOR PLAN COMPLEXITY AND SIGNAGE

To investigate the joint effects of plan complexity and signage on wayfinding, the results for each dependent variable were plotted (see Figures 3 through 6). For this analysis, the textual and graphic signage conditions were collapsed into one category, "Signage." Tests for significant interactions between floor plan complexity and signage were calculated as part of the two-way ANOVA computed for the previous analyses.

Figure 3 depicts the data for rate of travel across all settings for the two signage conditions. There is no significant interaction between floor plan complexity and signage conditions on rate of travel. Figure 3 shows that the presence of signage consistently increases rate of travel, regardless of level of plan complexity. However, rate of travel does not necessarily decrease in a linear fashion as plan complexity increases (also see Table 2).

Figure 4 depicts the data for backtracking behavior across all settings for the two signage conditions. This figure clearly shows the significant reduction in backtracking behavior between the signage and nonsignage conditions. There is no significant interaction between complexity of floor plan configuration and signage condition on backtracking. These results indicate that signage reduces backtracking errors regardless of the complexity of floor plan form.

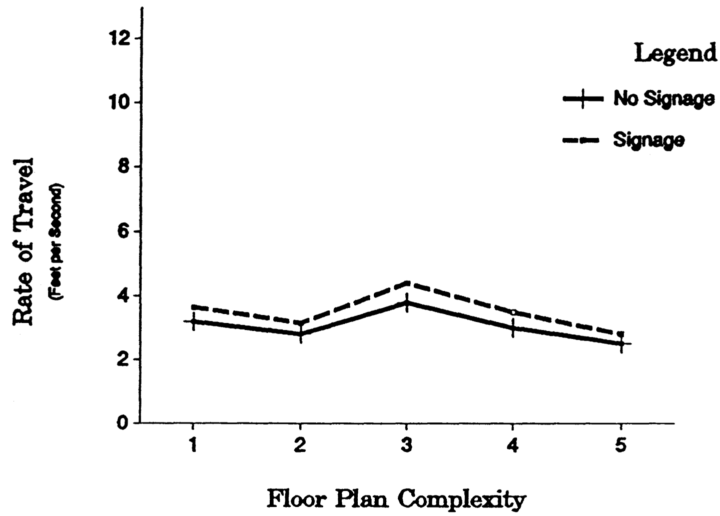

Figure 3: Effects of Floor Plan Complexity and Signage on Rate of Travel

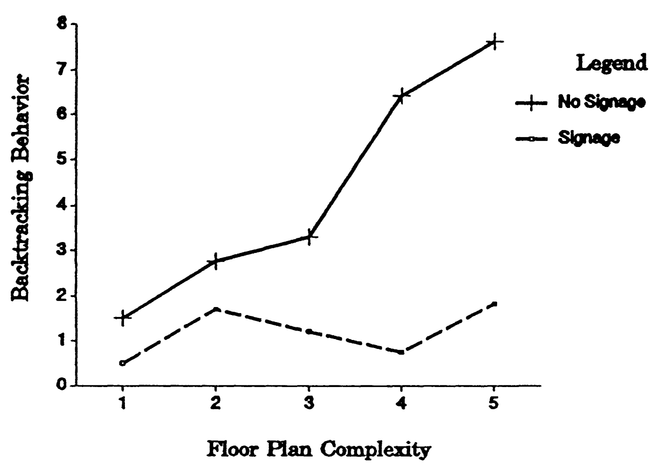

Figure 4: Effects of Floor Plan Complexity and Signage on Backtracking

Figure 5 illustrates the data for stopping-and-looking behavior across all settings for the two signage conditions. Although floor

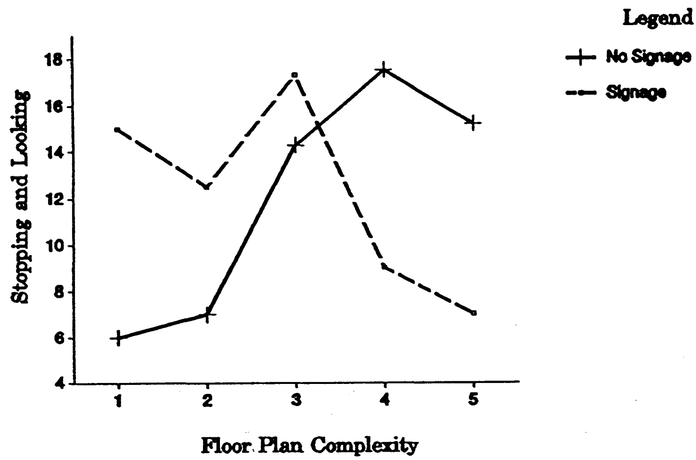

Figure 5: Effects of Floor Plan Complexity and Signage on Stopping and Looking

plan form significantly influenced stopping-and-looking behavior, the presence (or absence) of signage did not influence hesitations (see Tables 2 and 3). Figure 5 illustrates the significant interaction between the plan complexity and signage conditions ($F[8, 54] = 2.79, p < .05$) on hesitations. The data suggest that hesitations are not a problem in very simple settings with no signage, but that hesitations increase as complexity increases. However, the addition of signage is not a straightforward solution to hesitations. At low levels of plan complexity, signage appears to be more of a distracter than an aid, because simple settings with signage actually cause more hesitations than settings without signage. As floor plan complexity increases, signage becomes more effective in reducing hesitations.

Figure 6 illustrates the data for wrong turns across all settings for the two signage conditions. The results indicate that regardless of presence of signage, wrong turns do not become a significant problem until relatively high levels of plan complexity are reached (see Table 2). However, overall, people make fewer wrong turns in settings with signage than in those without (see Table 2).

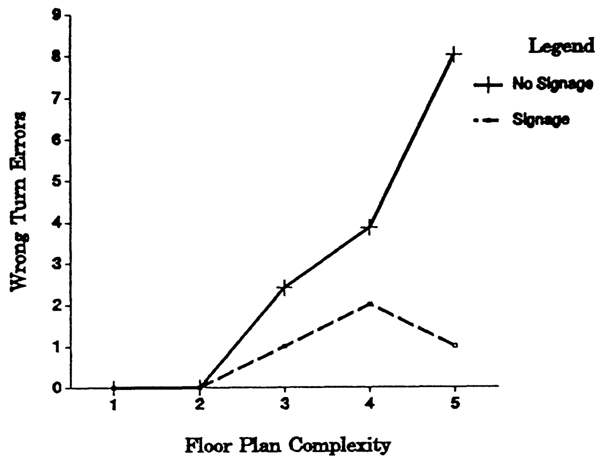

Figure 6: Effects of Floor Plan Complexity and Signage on Wrong Turn Errors

There is a significant interaction between the plan complexity and signage conditions ($F[8, 54] = 9.03$, $p < .001$) for wrong turn behavior (see Figure 6). Because the interaction term is significant, the data indicate that signage is not an effective means of reducing wrong turn errors in simple and moderately complex settings. Rather, floor plan configuration may have an overriding influence on wayfinding performance regardless of the presence of signage. This finding is congruent with Weisman's (1981) research, in which floor plan layout predicted 56% of the variance in reported wayfinding difficulty. Although signage does not increase wrong turn errors at lower or moderate levels of plan complexity, it does not reduce these errors. Similarly, signage does not reduce hesitations at lower or moderate levels of plan complexity for people with a modest amount of experience within the environment (see Figure 5).

CONCLUSIONS

As expected, the results show that an increase in plan complexity is related to a decrease in wayfinding performance. The addition of signage resulted in a 13% increase in rate of travel, a 50% decrease in wrong turns, and a 62% decrease in backtracking across the five settings. Graphic signage produced the greatest rate of travel in all settings, but textual signage was the most effective in reducing wayfinding errors, such as wrong turns and backtracking. These findings suggest that graphic and textual signage may be applied to optimize different aspects of wayfinding depending on the needs of the facility. Although wayfinding accuracy is always desirable, there are building types in which rate of travel is also an important consideration, and accuracy to a specific destination less critical. One such facility type is the transportation center (subway, bus, airport) through which large numbers of people must maneuver efficiently. Graphic signage might improve the overall flow of people through such a facility with minimal impact on accuracy. In hotels or institutional settings where emergency egress demands place a critical value on accuracy and speed, textual signage at key decision points combined with graphic signage for reassurance at intermediate locations could be used.

Despite the use of signage, plan configuration was found to exert a significant influence on wayfinding performance, because participants with access to signage in the most complex settings still made significantly more wrong turns than those in the simplest settings with no signage (see Figure 6). On other measures of wayfinding performance, such as rate of travel, backtracking, and hesitations, wayfinding was no better in the most complex settings with signage than it was in the simplest settings with no signage. Apparently, the presence of signage is not able to compensate for wayfinding problems due to the complexity of the floor plan.

This interaction between plan complexity and signage may be responsible for the lack of consensus on the effectiveness of signage found in the literature. In addition, interactions may occur between signage and variables such as visual access, or architectural features used as landmarks. The architectural elements that people use to orient themselves within buildings are thought to interact in as yet unspecified ways (Weisman, 1981). The results of this study reinforce the notion that multiple architectural variables and their interrelationships are a factor in understanding the influence of the designed environment on wayfinding performance.

Researchers have emphasized the importance of conceptualizing all features of the built environment as a wayfinding system (Hunt, 1985; Weisman, 1987). The present research demonstrates the value of using multiple architectural variables within the context of a field study, because complex relationships between variables were uncovered that might have otherwise gone undetected. The use of actual observations of wayfinding performance permits a clearer understanding of how design features influence human performance.

There are many forms of environmental information that the designer can use to help the building user understand and efficiently navigate through built space. The two types of information assessed by this study, floor plan form and signage, were found to interact in sometimes unexpected ways. The usefulness of signage, and different types of signage, varied with even minor differences in floor plan complexity. Various aspects of wayfinding performance are influenced differently depending on the combination of architectural cues available. Additional research needs to take a multivariate perspective so that the relationships between these and other sources of orientation information can be better understood.

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